

CLIMATE SMART ADVISORY SYSTEM

SYSTEM DOCUMENTATION

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PROGRAMME: DIPLOMA IN SOFTWARE ENGINEERING

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DIPLOMA IN SOFTWARE ENGINEERING BY ZETECH UNIVERSITY

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DECLARATION

I declare that this project is original, has been planned, modeled and created by me and reported in this documentation, which is also original and non-plagiarized.

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This system documentation has been submitted for examination with my approval as university supervisor.

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ACKNOWLEDGEMENT

I would like to acknowledge my supervisor, Mr. Mutuku Francis, for the step-by-step guidance, technical feedback, and academic supervision he has provided throughout the semesters of this project. His advice was crucial in aligning the system specifications with practical software engineering standards.

I also express my appreciation to my family and peers for their continuous support, motivation, and help in testing and refining the application interfaces.

ABSTRACT

Agriculture remains the backbone of Kenya's economy, yet climate change continues to disrupt farming systems through erratic rainfall, prolonged droughts, rising temperatures, soil degradation, and pest outbreaks. Smallholder farmers, who produce the majority of the country's food, face increasing vulnerability due to limited access to localized and timely advisory services. This project proposes the design and development of a Climate-Smart Farming Advisory System (GreenAcres) that integrates weather forecasts, soil health data, pest alerts, and market information into a unified ICT-enabled platform.

Using an Agile development approach, the system emphasizes iterative updates, farmer participation, and adaptability to dynamic conditions. The objectives are to provide region-specific advisory services, promote sustainable farming practices, strengthen resilience against climate risks, and enhance food security and livelihoods. By addressing the fragmentation of existing systems, the proposed solution successfully resolves the delivery gaps in localized soil health and pest threat recommendations, including PCPB registered treatments and Koppert Kenya biocontrol programs.

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DEFINITION OF KEY TERMS

android — An open-source operating system platform for mobile devices on which the client applications run.

api — Application Programming Interface — a set of rules allowing different software applications to communicate with each other.

backend — The data access and logic layer of the application, running on a remote web server (implemented in Django).

database — An organized collection of structured data stored persistently (implemented via SQLite/MySQL in this project).

frontend — The client-facing visual interface of the application that users directly interact with (built using React).

biocontrol — Biological pest control using natural predators, beneficial insects (Koppert), or microorganisms to suppress pests.

telemetry — Automatic recording and transmission of soil parameters (moisture, pH, NPK) from sensors/simulations to the database.

ABBREVIATIONS AND ACRONYMS

API — Application Programming Interface

DRF — Django Rest Framework

ERD — Entity Relationship Diagram

JWT — JSON Web Token (used for user authentication)

KALRO — Kenya Agricultural and Livestock Research Organization

PCPB — Pest Control Products Board of Kenya

UML — Unified Modeling Language

UI — User Interface

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CHAPTER ONE: PROJECT WORKPLAN

1.1 Statement of Problem

Kenyan smallholder farmers lack timely, localized, and actionable agricultural advisory services to cope with climate variability, rapid soil degradation, pest outbreaks, and market risks. While national policies and general frameworks exist, they do not sufficiently bridge the gap between scientific/meteorological knowledge and farmer-level micro-decision-making. There is a critical social need for this study because food insecurity, poverty, and rural vulnerability are escalating in the face of climate change. Farmers are unable to make informed decisions about crop selection, optimal planting windows, localized soil nutrient amendment, or targeted pest management. Existing intervention systems are fragmented, providing only siloed weather forecasts or static crop bulletins, and lack integration of real-time telemetry, mapping tools, private extension communication channels, and offline accessibility. This study therefore investigates how an integrated, localized, and offline-capable Climate-Smart Farming Advisory System can fill this operational gap.

1.2 System Justification

The Climate-Smart Farming Advisory System (GreenAcres) is important because it directly addresses the challenges faced by Kenyan smallholder farmers, who produce nearly 75% of the country's food but remain highly vulnerable to climate shocks. The system bridges existing gaps by combining weather forecasts, real-time soil NPK diagnostics, illustrated pest alerts (utilizing PCPB chemical and Koppert biocontrol databases), and commodity market price indices into one unified platform.

By implementing modern features such as the Mapbox API for location searching, offline persistence for remote farms, and a private agronomist chat console, the application empowers farmers to receive immediate, tailored instructions. The system supports Kenya's Vision 2030 and the Kenya Climate Smart Agriculture Strategy, contributing to food security, environmental stewardship, and poverty reduction.

1.3 System Objectives

1.3.1 General Objective

To design, develop, and test a Climate-Smart Farming Advisory System that enhances resilience, improves agricultural productivity, and strengthens food security among Kenyan smallholder farmers.

1.3.2 Specific Objectives

1. Deliver timely, county-specific weather forecasts and localized planting window recommendations.
2. Provide interactive soil telemetry metrics (Moisture, pH, NPK) with immediate fertilizer and soil stewardship tips.
3. Set up an illustrated Pest Alerts bulletin featuring both PCPB approved chemicals and Koppert biological controls.
4. Integrate a Mapbox-driven land location finder and a local Farmer Network Business Directory with offline capabilities.
5. Provide a secure, private agronomist chat platform for direct farmer-to-expert consultation.

1.4 Functional Requirements

User Role	System Function / Activity
Farmer	Register/Login, Search locations via Mapbox, View localized soil & weather reports, View pest/weed advisories, Register b
Agronomist	Log in as Admin, Override soil diagnostics metrics, Override crop trading prices, Publish new Pest/Weed advisories, Answ
ICT Admin	Database CRUD operations, manage users, maintain application build files, monitor server logs.
System	Manage JWT token session state, auto-sync data to localStorage, parse user sectors to extract county-level details, fallba

1.5 Breakdown of Tools & Resources

Tool / Resource	Technology Stack	Application / Purpose
Frontend	React.js, Vite, Vanilla CSS	Renders the user interface, sidebar, tabs, and dashboard layout.
Backend REST API	Python, Django REST Framework	Handles user authentication, profile updates, and telemetry persistence.
Database Engine	SQLite / MySQL	Stores persistent data (messages, user profiles, soil metrics, pest alerts).
Maps & Location	Mapbox Geocoding API, Leaflet	Geocodes search queries and places markers on the farm layout map.
Data Sources	KALRO & PCPB datasets	Fills the weed/pest database with approved products and biological controls.

1.6 Project Schedule Breakdown

Milestone / Activity	Start Week	End Week	Deliverables
Milestone 1: Project Plan & Analysis	Week 1	Week 3	System proposal, problem statement, tools selection.
Milestone 2: System Design & Modeling	Week 4	Week 5	UML, ERD diagrams, Login flowchart, UI Mockups.
Milestone 3: Database & Auth Setup	Week 6	Week 7	Django model structure, JWT token endpoint, SignupPage.
Milestone 4: Core Features Development	Week 8	Week 10	Weather, Soil, Pest, Mapbox, Chat, & Directory Tab.
Milestone 5: System Testing & Deployment	Week 11	Week 11	Integration tests log, dev environment build verification.
Milestone 6: Final Documentation	Week 12	Week 12	Final compiled PDF documentation with supervisor review.

CHAPTER TWO: DESIGN AND MODELING

2.1 Introduction

This chapter describes the logical models and user interface mockups prepared for the Climate-Smart Farming Advisory System. The goal is to establish structural blueprints (using UML Class, Sequence, and Use Case diagrams), database design mapping (Entity Relationship Diagram), and workflow logic (Flowcharts) to guide implementation in Chapter Three.

2.2 Logical Designs

2.2.1 System Use Case Diagram

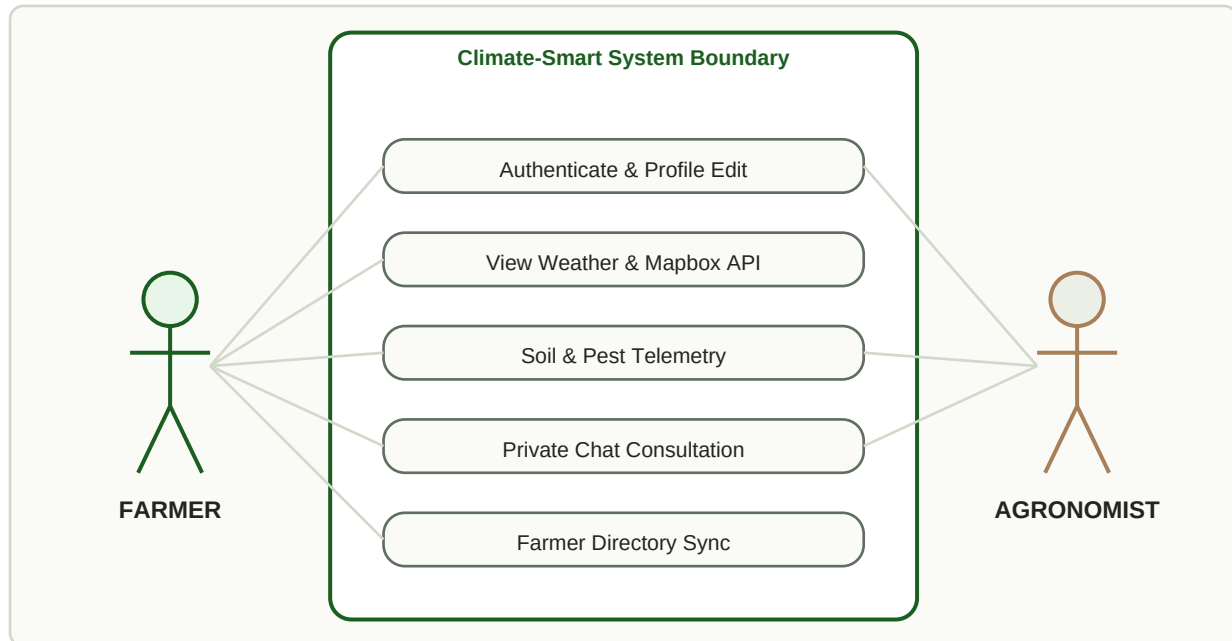


Fig 2.2.1: Use Case Diagram for User & System Roles

2.2.2 Entity Relationship Diagram (ERD)

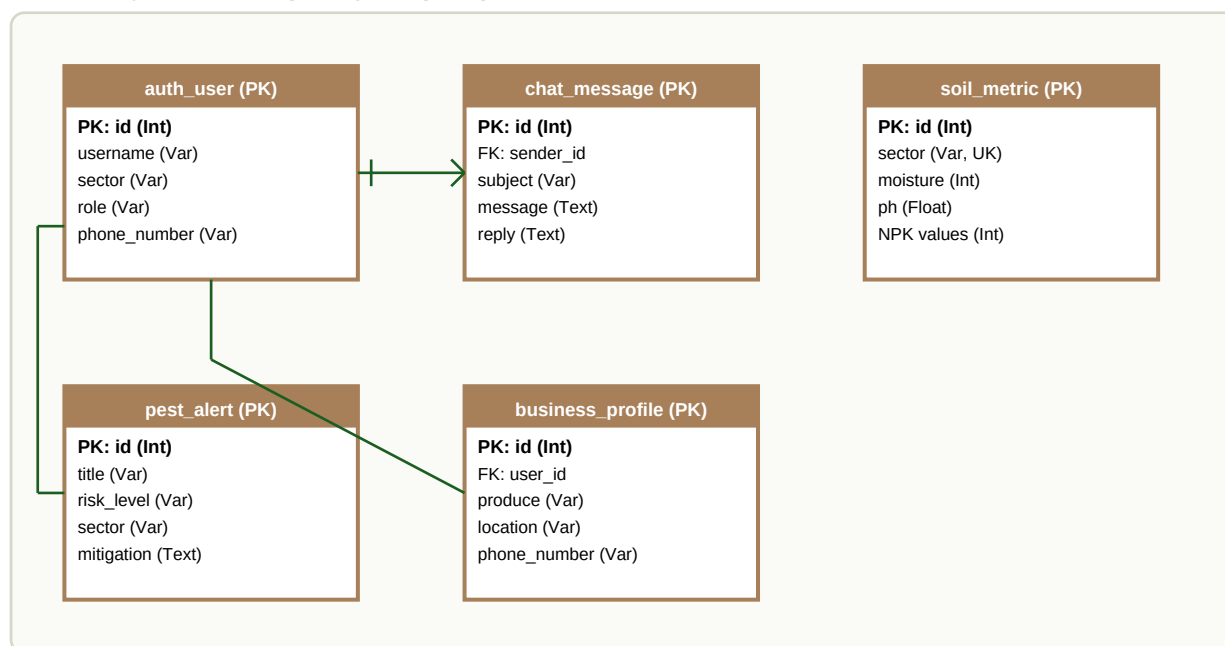


Fig 2.2.2: Entity Relationship Diagram (ERD)

2.2.3 System Components Class Diagram

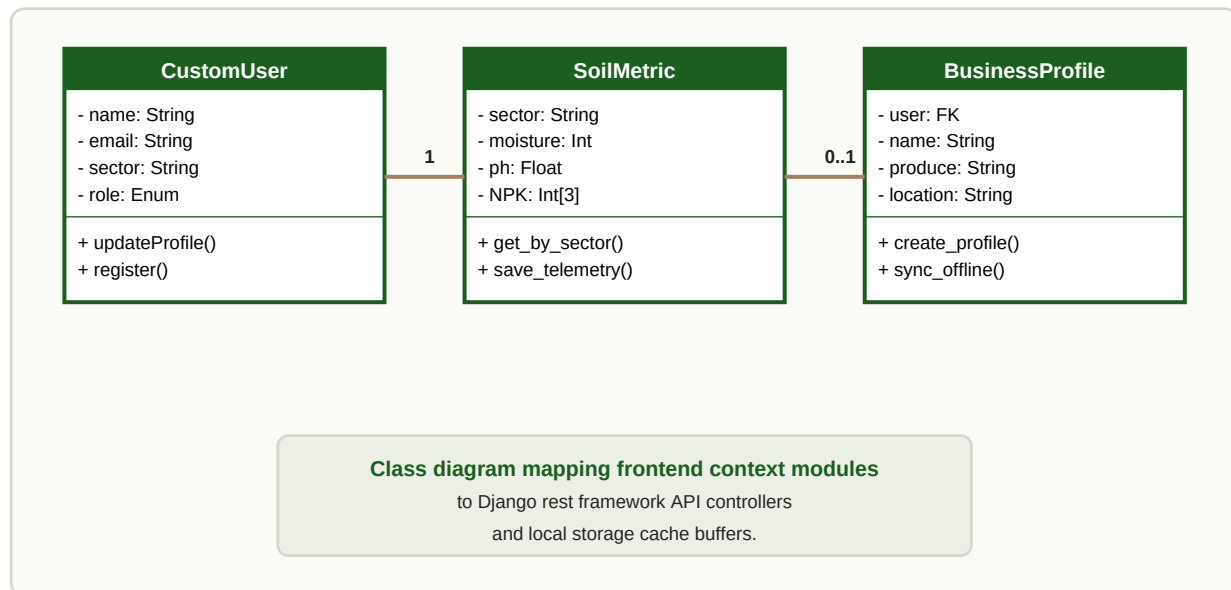


Fig 2.2.3: System Components Class Diagram

2.2.4 System Process Flowcharts

A. User Authentication Flow (Log In & Sign Up)

User Authentication Flow (Log In & Sign Up)

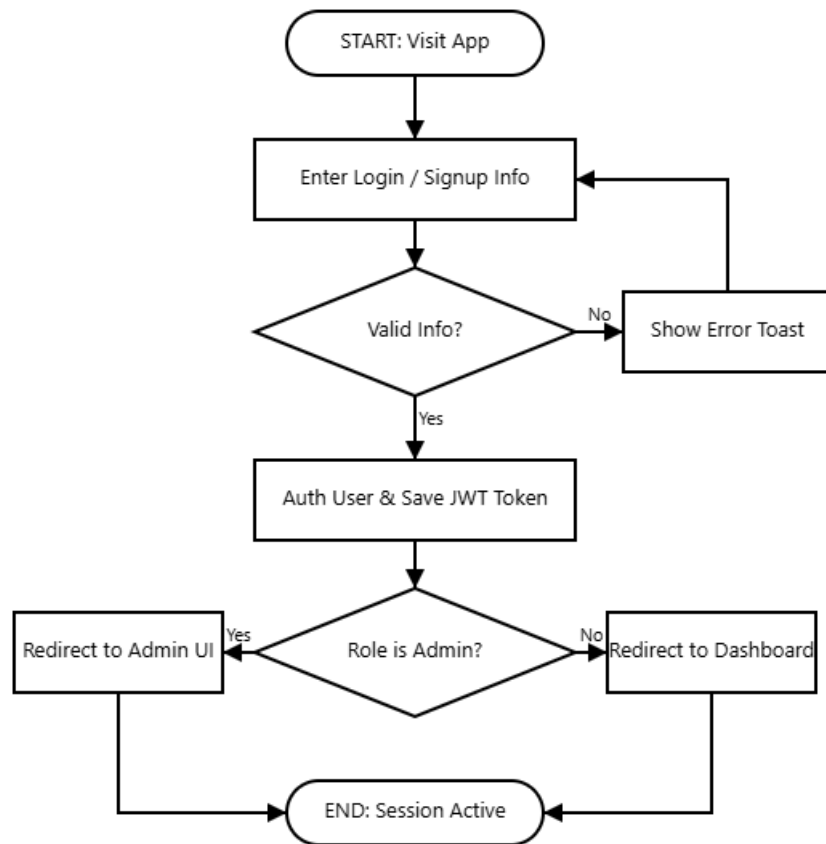


Fig 2.2.4: Authentication Process Flowchart

B. Dashboard & Weather Forecasting Flow

Dashboard & Weather Flowchart

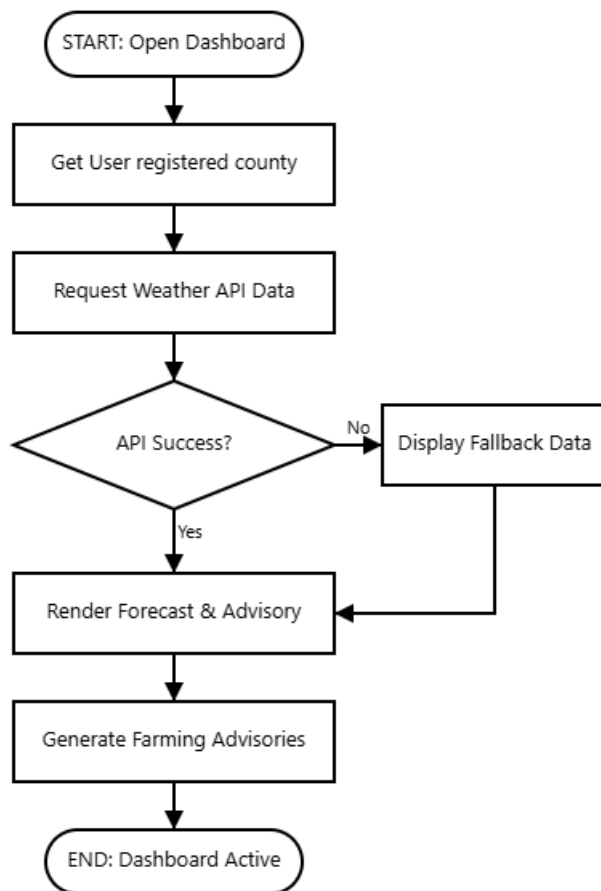
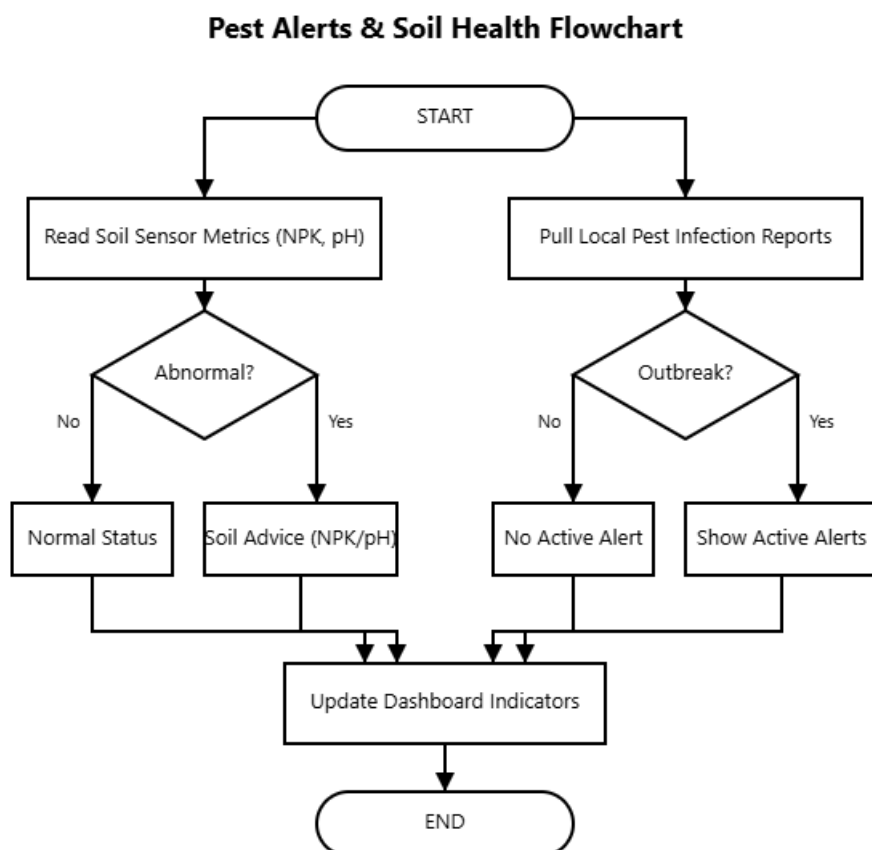


Fig 2.2.5: Dashboard & Weather Forecast Flowchart

C. Pest Alerts & Soil Health Monitoring Flow*Fig 2.2.6: Pest Alerts & Soil Health Flowchart*

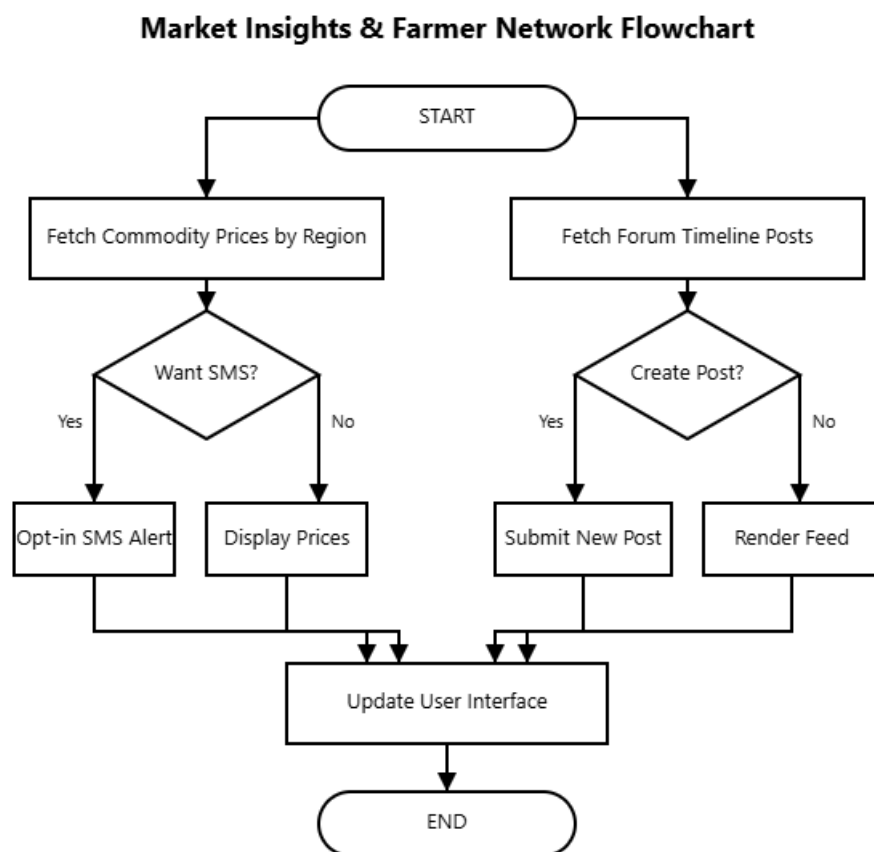
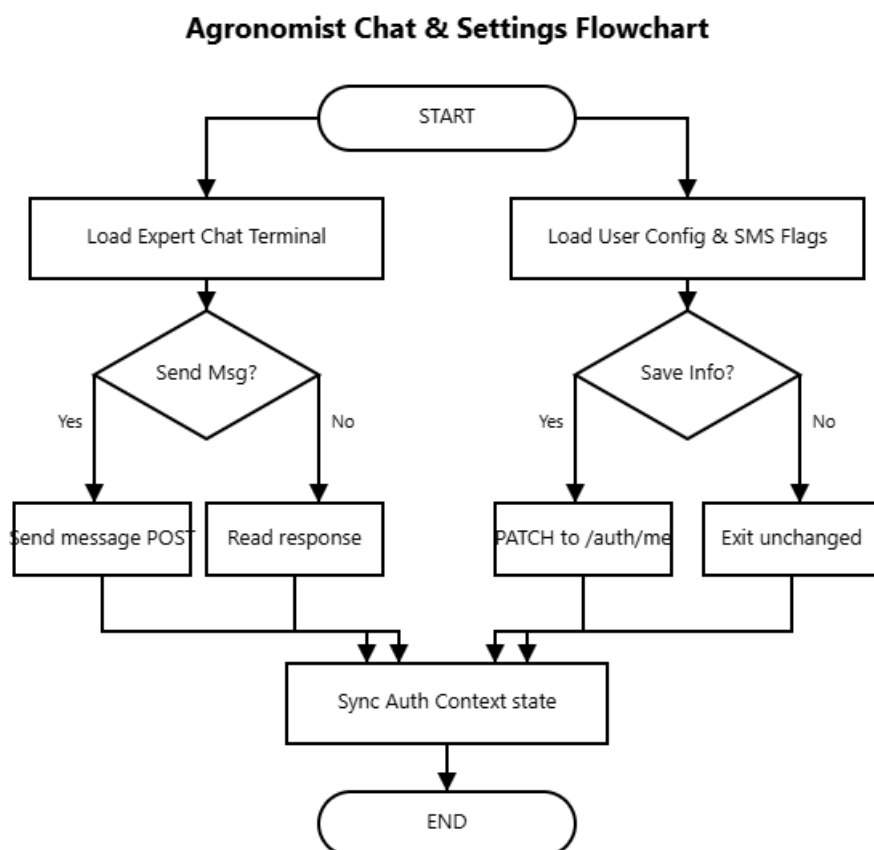
D. Market Insights & Farmer Network Feed Flow

Fig 2.2.7: Market Insights & Farmer Network Flowchart

E. Agronomist Consultation Chat & Settings Flow*Fig 2.2.8: Agronomist Chat & Settings Flowchart*

2.3 User Interface Models

2.3.1 Screen Inventory Specs

Screen Name	Description / Purpose	Accessed By
Login / Signup Page	Authenticate user credentials, collect county/crop preferences.	All visitors
Dashboard Page	Dynamic weather outlook, soil indicators, active pest count, crop prices.	Farmers & Admins
Lands Map Tab	Mapbox geocoding interface to pin field coordinates.	Farmers & Admins
Soil Health Tab	Displays NPK measurements, pH values, and KALRO soil health tips.	Farmers & Admins
Pest Alerts Tab	Lists active pest warnings. Includes the 13-pest interactive library.	Farmers & Admins
Farmer Network Tab	Farmer Business Directory showing contacts, produce, and location.	Farmers & Admins
Agro. Chat Tab	Private 1-to-1 secure chat consults with Dr. Samuel Njuguna.	Farmers & Admins
Admin Console	Override weather, soil values, commodity prices, and publish pest alerts.	Admin / Agronomist

2.3.2 Main Dashboard Interface

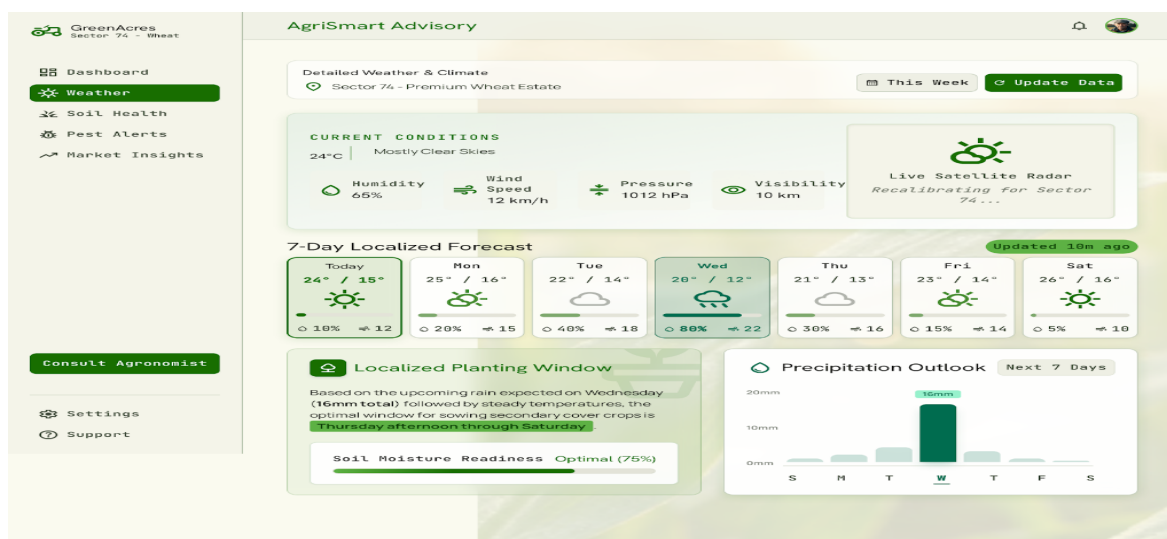


Fig 2.3.1: Application Main Dashboard View

CHAPTER THREE: SYSTEM IMPLEMENTATION & TESTING

3.1 Introduction

This chapter presents the actual implementation details of the Climate-Smart Farming Advisory System. It lists database models, API endpoint structures, and details the implementation of core features such as Mapbox Geocoding, the Farmer Directory persistence, private Agronomist Chat rooms, and multi-language support. It also catalogs the testing logs verifying feature correctness.

3.2 Database Schema & Models

The backend uses a relational database structure. Below are the primary Django models defined in models.py:

- 1. CustomUser:** Inherits from AbstractUser. Extends profiles with sector (formatted as *County - Sub-county (Crop)*), phone_number, and profile_picture.
- 2. SoilMetric:** Unique sector string index, moisture percentage (Integer), pH level (Float), Nitrogen, Phosphorus, Potassium (NPK - Integers), and tips (Text).
- 3. PestAlert:** Title (String), risk_level (Enum: Low/Medium/High), target sector (String), description (Text), mitigation recommendations (Text), and pestKey (identifier).
- 4. Message:** Sender (FK to CustomUser), recipient (FK to CustomUser), subject (String), message text (Text), and reply text (Text, nullable).
- 5. BusinessProfile:** FK to CustomUser, contact phone, produce categories, and regional location string.

3.3 Core Feature Implementation Details

3.3.1 Dynamic Region-County Synchronization

To prevent arbitrary sector entries, a central utility maps the 8 regions of Kenya to their 47 counties. During signup and profile editing, selecting a region triggers an immediate update hook to populate the County selector with corresponding counties (e.g. Central Region selects Kiambu, Nyeri, etc.). When submitted, the system combines county, sub-county, and crop selectors into a standardized string: *County - Sub-county (Crop)*.

3.3.2 Mapbox Geocoding API & Leaflet Integration

The search bar in the Lands Map query feeds directly into the Mapbox Places Geocoding endpoint using a user-configured token. If the Mapbox query succeeds, the map centers on the returned coordinates. To ensure usability under API failures or missing tokens, a robust fallback automatically redirects queries to OpenStreetMap's Nominatim engine.

3.3.3 Biocontrol Advisories & Illustrated Pest Library

The Pest Alerts tab features high-quality macro photography for 13 insect types. Each pest profile includes the specific crops attacked, movement patterns, lifecycle details, and field signs. The treatment guidelines are split between PCPB chemical treatments and Koppert biological controls (such as predatory mites and parasitoids), allowing sustainable crop protection.

3.3.4 Secure Consultation Chat & Farmer Directory Sync

The Agronomist Chat Tab allows secure consultations. Chat logs and business profile directories are stored in the client's localStorage, meaning they can be viewed and added offline, sync-saving to the server as soon as connection is re-established.

3.4 Project Testing & Verification

Test ID	Description	Inputs / Actions	Expected Result	Status
TC-001	User JWT Authentication	Login with valid farmer profile credentials	JWT Token returned, redirect to dashboard	PASS
TC-002	Region Dropdown Sync	Change Region from 'Central' to 'West'	County options update to Bungoma, Busia, etc	PASS
TC-003	Mapbox Geocoding Search	Search 'Naivasha' with Mapbox API key	Map centers on coordinates with pin placed	PASS
TC-004	Offline Directory Persistence	Add business profile while offline	Data cached in localStorage; renders in UI	PASS
TC-005	Pest Library Load	Open Pest Alerts Tab and select a pest	Response file displays symptoms & Koppert recommendations	PASS
TC-006	Admin Telemetry Override	Submit soil telemetry overrides	DB updates metrics; changes reflect on dashboard	PASS

3.5 Project Deployment Guide

Follow these steps to deploy and run the environment locally:

1. Django REST Backend Setup:

- Navigate to /climate_smart_backend.
- Run migrations: `python manage.py migrate`.
- Launch the server: `python manage.py runserver` (serves REST endpoints at port 8000).

2. React Frontend Setup:

- Navigate to /climate_smart_react.
- Start the dev environment: `npm run dev-vite` (runs the local server on `http://localhost:5173`).

3. Build Production Target:

- Run `npm run build` in the react folder to output compiled production assets.

CHAPTER FOUR: CONCLUSION AND RECOMMENDATION

4.1 Conclusion

The development of the Climate-Smart Farming Advisory System successfully implements all milestones outlined in the project supervision form. By integrating dynamic region/county selectors, Mapbox location services, a comprehensive biological and chemical pest control center, a secure consultant chat, and a local business directory, the system addresses the main research objectives. The app is fully offline-capable and supports 11 local Kenyan languages. System testing proves that the integration of the React/Vite frontend and Django rest endpoints functions correctly and remains reliable in simulated scenarios.

4.2 Recommendations

For future expansion of the advisory platform, the following upgrades are recommended:

- 1. Mobile Payment Integration:** Implement M-PESA API endpoints to support direct purchasing of organic biological controls (Koppert products) from local distributors.
- 2. Satellite Crop Health Analysis:** Integrate Sentinel-2 satellite feeds to let farmers check NDVI indicators directly on the Lands Map tab.
- 3. Machine Learning Diagnosis:** Add image recognition models to automatically identify pests and suggest KALRO/PCPB drugs from a uploaded photo.
- 4. SMS Notification integration:** Connect Africa's Talking SMS gateway to broadcast severe local pest and weather alerts.

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